

Flexural Modeling and Design of FRC, TRC, and UHPC using Serviceability and Sustainability based Criteria

Barzin Mobasher

^a School of Sustainable Engineering and Built Environment, Arizona State University, Tempe, AZ, USA

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Abstract

An overview of the modeling work on the structural simulation and design of FRC, TRC, and UHPC reinforced concrete materials is presented. Three main design parameters consist of the mechanical properties of the materials, the section parameters in terms of size, geometry, type and placement of reinforcement in discrete or distributed systems, as well as the structural redundancy factors in terms of loads and imposed boundary conditions that affect the response. The components of non-linear material composite response of strain-hardening and softening is addressed through the homogenization from a microstructural level to the macro level. In order to correlate the material ductility to structural ductility, traditional ideas of localization by a quasi-plastic yield hinge and yield line analysis are used. The modes that contribute to the ductility under varying loading combinations relate the reinforcement and matrix's interaction, and the role of reinforcing mechanisms of ductility through cracking and crack distribution.

Development of the design tools are based on the generation of closed form solutions for the moment-curvature response of a section of arbitrary shape with different tensile and compression constitutive law, and reinforcement position and properties. The design equations are developed by methodic calculation of stress distribution, deflection, curvature, with the parametric composite model, while utilizing the nonlinear response in the strain hardening and softening. Generalized relationships of strain and stress across the section are integrated into internal forces, moment, and curvature relations using the material and geometrical properties in closed form. These derivations are evaluated to point out the effect of uniform and varying strain fields, interface mechanics, flexural toughening and movement of neutral axis, and imposing shear failure criteria. Using a simplified moment-curvature response, the load deflection distribution along the length of a beam can also be obtained. Guidelines for the development of tools for specialized applications for hybrid reinforced concrete systems, TRC, UHPC, and sandwich composites are presented and applied to classic reinforced concrete sections. Specialized cases of bridge deck slabs and girders, water retaining structural panels, light-rail track slabs, and segmental tunnel linings are simulated and compared with the experimental data. The generalized models address the flexural design using parameters such as maximum allowable curvature, crack width, deflection, or any other serviceability-based criteria through the parameterized moment-curvature relationship. A variety of sustainable design guidelines for new applications and redesign of hybrid systems, minimum reinforcement, maximum crack with limits as well as stiffness and deflection-based categories are possible. Using this approach, the selection of both the material and structural designs are concurrently accomplished.